



Institute of Engineering and Technology (IET)

Major Project Synopsis(PR1107)

NeetiNex : An AI-Powered Government Scheme Assistance Platform

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Table Of Content

Table Of Content	2
1. List of Figure	2
2. List of Abbreviation	3
3. Abstract	4
4. Introduction	5
5. Problem Statement	6
6. Objectives	7
7. Scope of Work	8
8. Methodology	9
9. System Architecture	11
10. Fault Tolerance & Reliability	12
11. Expected Outcomes	13
12. References	14

List of Figure

Fig. No	Figure Name	Page No.
1	RAG Pipeline	10
2	System Architecture	11

List of Abbreviations

S. No	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	RAG	Retrieval-Augmented Generation
3	LLM	Large Language Model
4	NLP	Natural Language Processing
5	STT	Speech-to-Text
6	TTS	Text-to-Speech
7	API	Application Programming Interface
8	PDF	Portable Document Format
9	FAQ	Frequently Asked Questions
10	UI	User Interface
11	DB	Database
12	BM25	Best Matching 25
13	URL	Uniform Resource Locator
14	OCR	Optical Character Recognition
15	HTTP	Hypertext Transfer Protocol
16	JSOB	JavaScript Object Notation

1. Abstract

In this project an AI based Government Scheme Assistant will be developed to assist the citizens in finding and comprehending government schemes. Users do not have to specify a particular scheme, department or application process. They can just say what they want and the system will offer them appropriate schemes. Users can also get detailed information about a particular scheme including purpose of the scheme, eligibility, benefits, links, documents required for applying, application process, FAQ's, Sources etc.

Initially, Central Government and Rajasthan Government schemes will be included in the system. Official Government websites, Ministry or Department websites, scheme guidelines, notifications and official PDF documents will be used to develop a trusted knowledge base. Before using the AI system the information will be collected, cleaned, organized, checked, and linked to the source of the information.

The system will employ Retrieval Augmented Generation (RAG), hybrid search, reranking, eligibility rules and conversational AI. A separate eligibility module will validate the user's information with the actual conditions of eligibility for each scheme. This will minimize the risk of providing incorrect eligibility details solely due to relying on the AI model. The system will also indicate the source of the information for an answer and will explicitly state if the information sought is not found in the official documents provided.

Access to the system will be from different means and the same backend will be utilized. This will encompass a dashboard on the web for all text and voice communications, a feature to upload documents to assist users in understanding government scheme documents, as well as a telephone interface that users can interact with through conversation - this will be an advanced feature.

Search accuracy, eligibility accuracy, answer accuracy, hallucination rate, and response time will be used to test the system.

2. Introduction

There are various schemes offered by government departments regarding education, agriculture, health, employment, housing, pension, women welfare, financial assistance and others. Though there are official government sites providing this information, it may not be easy for citizens to find the right scheme. Data is frequently distributed on various websites, pages, guidelines, notifications, and pdf documents.

A citizen may have an idea of the type of support they want financial support in education, help for agriculture, pension support or housing benefits but may not be aware of the scheme's name. However, a citizen might already know the name of a scheme but have queries regarding eligibility, benefits, documents required, and the application procedure.

Both situations will be addressed in the proposed system via one conversational interface. Users can articulate their need in simple terms, provide follow up questions as they see fit, or make a specific question about a particular scheme. The System will query a trusted government knowledge base and return an answer in a comprehensible manner with the original source.

The system does not aim to replace official portals or government departments but to give information and guidance. It will give information according to the available official criteria and clearly indicate when the final eligibility decision should be made by the competent government department.

3. Problem Statement

It can be challenging for citizens to locate and comprehend information about government schemes. They might not be familiar with a scheme's name, the eligibility criteria might be unclear, or they may not understand the necessary paperwork and application procedure. The information is also disseminated into various official websites, government portals, notifications, guidelines and PDF etc. Users asking questions in a conversational manner, asking follow up questions, or needing a simple explanation of a complex government document do not work well with a normal keyword-based search system.

Accessibility is another issue. Certain consumers might need to inquire about a certain matter by calling rather than via a website. Other users might want to access a dashboard to search for schemes, communicate with the assistant, voice input, upload a document issued by the government and ask questions about it. This means that one approach to access is not sufficient, due to these various needs.

The proposed project will address these issues with one Government Scheme Assistant, by means of three modes of access.

1. Web dashboard for searching schemes, conversing with the assistant and text/voice input.
2. An interface that takes the form of a telephone whereby users may ask questions and communicate with the assistant over the telephone.
3. An upload of document option that enables the user to upload an official government scheme document and pose questions about the document.

All three interfaces will share main AI backend, government knowledge base, RAG pipeline and eligibility engine.

The system will enable the user to state his problem and locate appropriate schemes, ask any questions about a known scheme, obtain information about benefits, eligibility, the documents that are required and the procedures involved in the application process, or be given the opportunity to upload a government document, and be given an explanation.

The system will be based on providing information from trusted government sources. It will seek to limit the number of wrong or unsupported responses and will give source references of the information with the response. When the necessary information cannot be determined from the documents in the system, it will explicitly state it, rather than providing a fabricated answer.

4. Objectives

- To create an AI-based conversational assistant which aids users in discovering and comprehending government schemes.
- To enable the users to ask questions about the government schemes through simple natural language chat and voice.
- To assist the users in identifying appropriate government schemes according to their requirement and eligibility.
- To provide detailed information about the schemes such as benefits, eligibility, documents required, application process and official sources.
- To gather and compile information on government schemes from official verified government resources.
- To create a system called Retrieval Augmented Generation (RAG) that generates answers from authoritative government information.
- To provide a module to check eligibility that can be compared to eligibility rules of a scheme.
- To enable users to upload government scheme documents via the dashboard and get queries related to them answered.
- To add references from official sources and minimise wrong or unverifiable AI-generated answers.
- To create one dashboard with features like uploading documents, chatting, voice interaction, and searching the scheme.
- To build a voice-controlled interface for the phone with the same AI functions as the web dashboard.
- To test and assess the system based on retrieval accuracy, eligibility accuracy, answer relevance, faithfulness and overall reliability.

5. Scope of Work

- Initially the system will be implemented for Central Government schemes and Rajasthan Government schemes.
- The initial knowledge base will contain around 50-100 verified schemes.
- The scheme information will include:
 - Scheme description
 - Benefits
 - Eligibility criteria
 - Required documents
 - Application process
 - FAQs
- Users will be able to search for schemes without being aware of the exact scheme name.
- Users will also be able to converse with great detail with a scheme they are already familiar with.
- If necessary, the system will ask further question when a user is missing some information.
- An eligibility module will be provided to offer potential eligibility information from the scheme criteria available.
- The web dashboard will contain the following:
 - Chat interface
 - Voice interaction
 - Scheme search
 - Document upload
- The users will be able to upload an official scheme document and ask questions on it or request a simple explanation.
- Users will be able to use the same AI assistant without having to open the dashboard with a separate phone-based calling interface.
- The system will primarily be based on official government websites, portals, documents and other government verified sources as a knowledge base.
- The system will be designed to allow for additional schemes and other states to be added in the future.

6. Methodology

6.1 Data Collection and Verification

The first step will be to develop a dependable database of government schemes. The primary source of information on the schemes will be the official websites of the ministry and department, government portals, official scheme guidelines, notifications, and government PDF documents. Each scheme will record information on the source including the source URL, document name, relevant page/section and date of last check.

Eligibility, benefits, necessary documents, application process, etc. will be verified with the original government source for important information. The system will separate verified information from information that needs more clarification and information that could not be found in the available official sources.

6.2 Data Processing

These web pages and PDF documents will be subjected to extraction and cleaning process. The details will be segmented into informative areas like scheme details, eligibility, benefits, documents required, procedure of application, FAQs, etc. This information will then be broken down into smaller pieces to be easily searched. Important scheme details will also be kept separately to be used for eligibility checking.

6.3 Knowledge Base and RAG

The information will be embedded in a suitable sentence transformer model and saved in a vector database (Chroma/FAISS). BM25 is going to be used to combine with vector search, enhancing search results. The results retrieved will be sent to a reranker which will determine which results are the most useful and then pass that information to the LLM.

The main process will be:

The structure of the information flow includes the following stages: Official Sources, Extraction, cleaning, chunking, embeddings, hybrid search, reranking, relevant information, LLM response.

Every chunk will retain information like scheme ID, section name, source URL and document or page number. This will assist the system in giving you the source and answer.

6.4 Conversational Profile and Eligibility.

The system will gather key information from a user while describing their condition and save the information in a temporary user profile. These details may contain information such as age, state, income, occupation, education, gender, and other information that may be required for a specific scheme. If any of the necessary info is missing, then the system will prompt the user with another question.

The rules, developed from the verified scheme criteria, will be used to verify eligibility. The LLM will be able to comprehend what the user is saying, and explain the result; the rule-based system will take care of the eligibility conditions that can be verified directly.

Possible outcomes include potentially eligible, not eligible (due to criteria limitations), more information needed, and unable to verify.

6.5 Dashboard and Document Understanding

A web dashboard that will allow searching for schemes, communicating with others, speaking with others, and uploading of document(s). Users will be able to upload an official government scheme pdf and ask questions like - What are the eligibility criteria? What are the documents required? Explain this document in Hindi? etc.

A separate temporary document RAG system will be used to process the uploaded document. It won't be included in the main government scheme knowledge base directly.

6.6 Calling and Voice Interface

The same main AI backend will be used for voice interaction. The user's speech will be first translated to text. This text will then pass through the conversation, the RAG system and the eligibility systems. The final answer will be converted back into speech.

The web dashboard will be developed first with a voice feature. The availability of the calling or telephony set up for the project will determine whether this advanced feature is added later as a telephone-based calling feature.

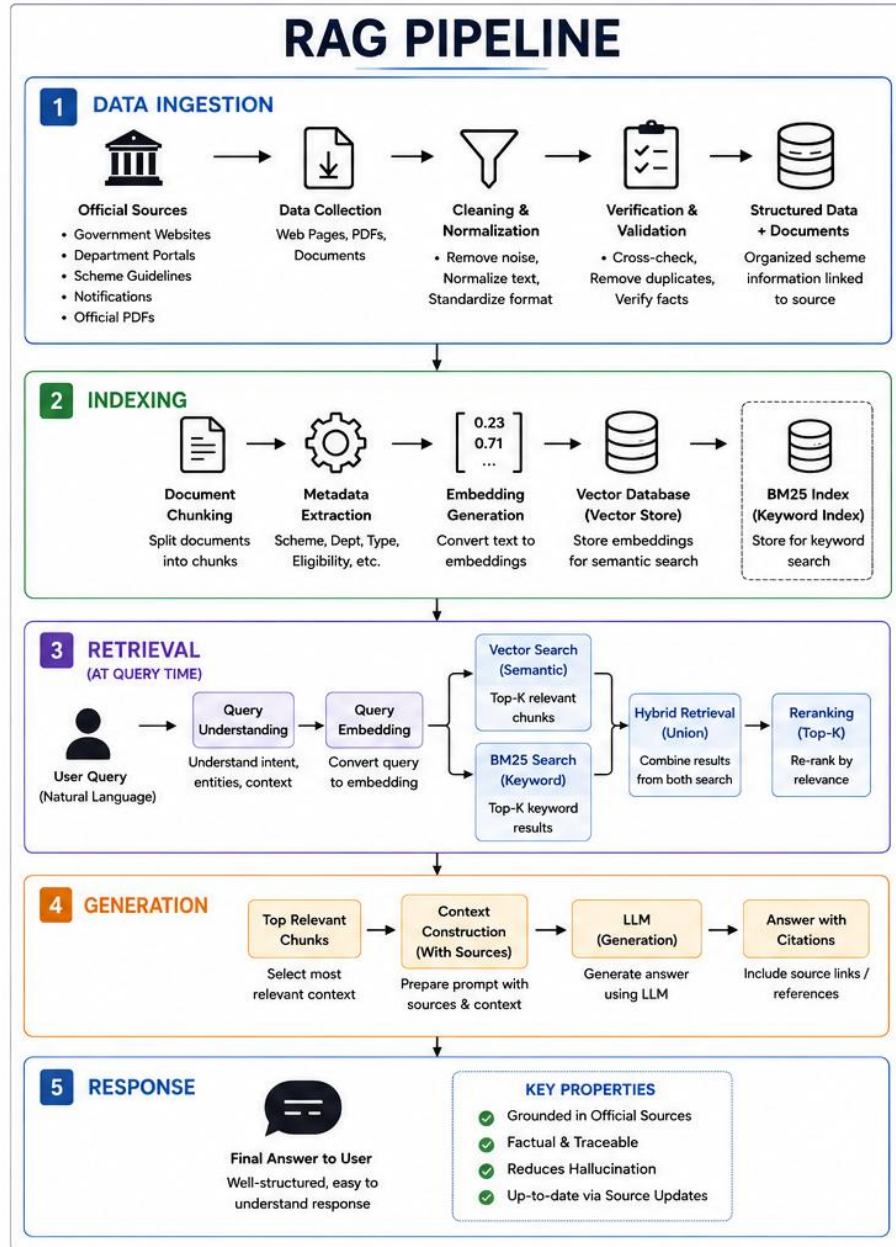


Fig. 1 RAG Pipeline

System Architecture:

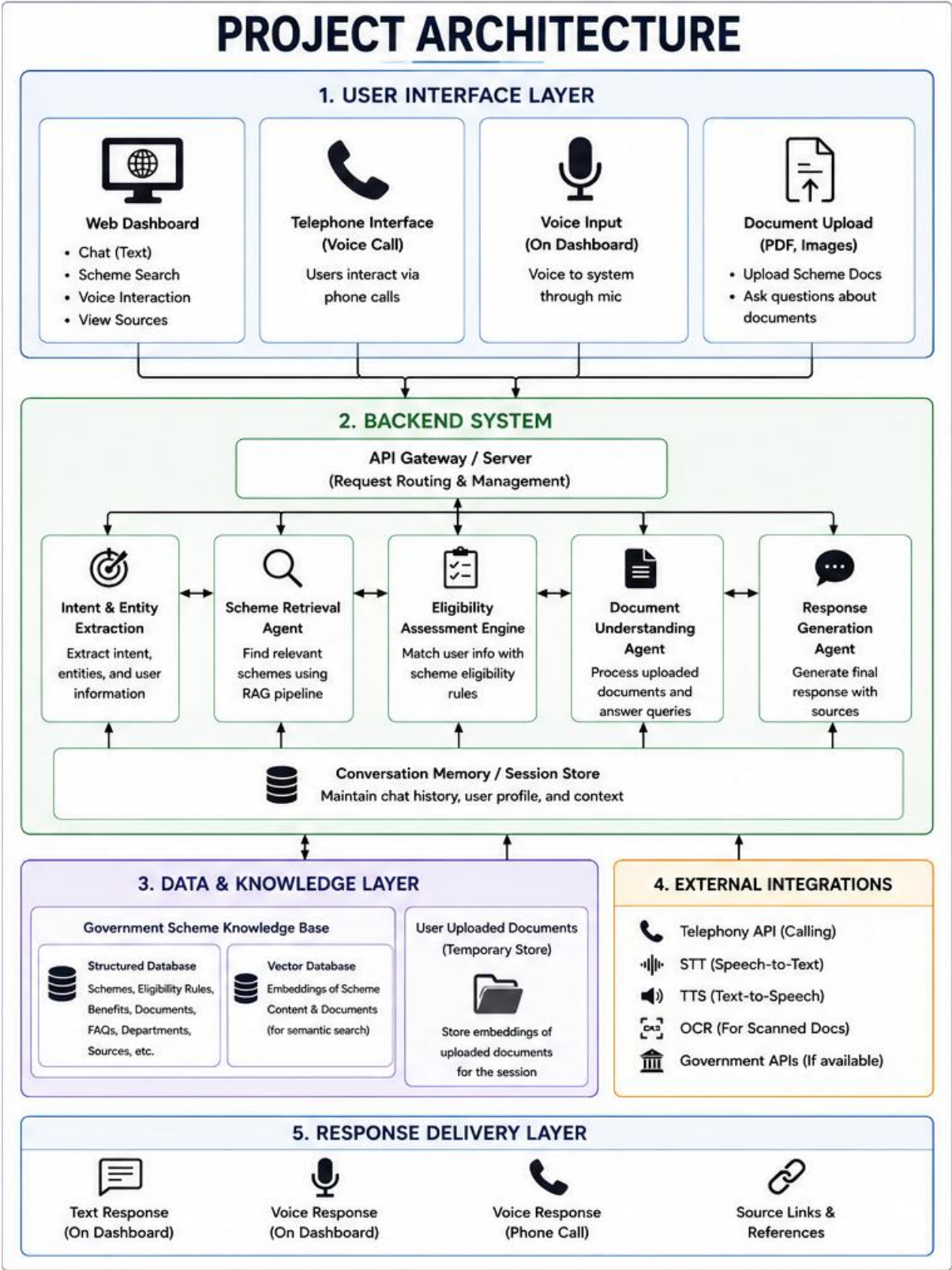


Fig. 2: System Architecture

7. Fault Tolerance & Reliability

- Information on important aspects of the scheme will be obtained from the sources that are verified by the official government.
- Wherever possible, the system will record source URLs and document/page information.
- Ability to produce answers based on retrieved and verified information rather than solely on the LLM's own knowledge.
- No Answer Handling: If the information is not available in the knowledge base, the system will state that the information could not be verified.
- Where possible eligibility will be validated according to the real scheme criteria rather than relying on the LLM.
- The relevant information regarding the scheme for instance, benefits, income thresholds, age restrictions, and the required documents will be verified against the original government source.
- User uploaded documents will be isolated and processed separately, and will not be added to the permanent government scheme knowledge base.
- Input validation: If the user fails to enter some information, the system will prompt the user with further questions rather than making assumptions.
- When voice service or any other external service is temporarily no longer operational, the text-based dashboard will remain functional if possible.
- Information Freshness: Each scheme record will have a verification date or an update date which will allow information to be identified as stale and re-checked.
- These will help improving the reliability and verification of the system. They'll also make sure that the AI doesn't report non-official information as official government announcements.

8. Expected Outcomes

At the end of the project, we will provide:

- A working prototype of a conversational Government Scheme Assistant.
- A verified knowledge base with around 50-100 Central Government and Rajasthan Government schemes.
- Official government websites and PDF documents: A data collection and verification pipeline.
- A hybrid approach to RAG based on vector search, keyword search, and reranking for relevance.
- A structured eligibility engine to perform scheme specific eligibility checks.
- A conversational interface which asks follow-up questions and has the ability to answer questions about government schemes.
- A web dashboard with text interaction, voice interaction, scheme search and document understanding capabilities.
- A document RAG feature to upload the official documents of a scheme and pose questions about them.
- A prototype of a telephone-based calling system if feasible within project time.
- Information is provided in answer to the questions placed based on government official information, where possible, with source references.
- A retrieval performance report, eligibility accuracy report, answer accuracy report, hallucination rate report and response time report.
- A working prototype, project report, presentation and final documentation of the system.

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